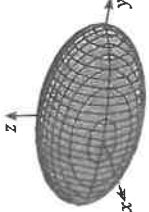
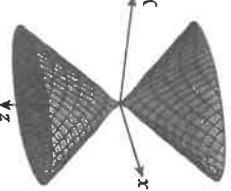
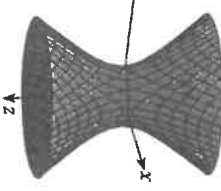
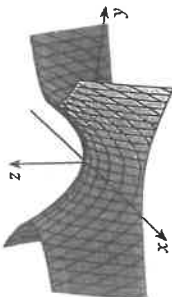
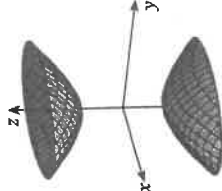
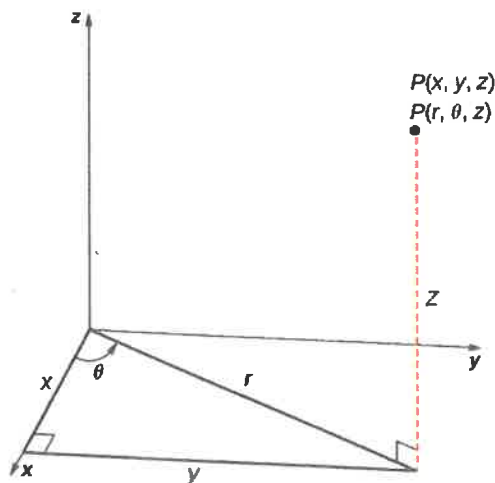


Table 1 Graphs of Quadric Surfaces

Surface	Equation	Surface	Equation
Ellipsoid 	$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$ All traces are ellipses. If $a = b = c$, the ellipsoid is a sphere.	Cone 	$\frac{z^2}{c^2} = \frac{x^2}{a^2} + \frac{y^2}{b^2}$ Horizontal traces are ellipses. Vertical traces in the planes $x = k$ and $y = k$ are hyperbolas if $k \neq 0$ but are pairs of lines if $k = 0$.
Elliptic Paraboloid 	$\frac{z}{c} = \frac{x^2}{a^2} + \frac{y^2}{b^2}$ Horizontal traces are ellipses. Vertical traces are parabolas. The variable raised to the first power indicates the axis of the paraboloid.	Hyperboloid of One Sheet 	$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1$ Horizontal traces are ellipses. Vertical traces are hyperbolas. The axis of symmetry corresponds to the variable whose coefficient is negative.
Hyperbolic Paraboloid 	$\frac{z}{c} = \frac{x^2}{a^2} - \frac{y^2}{b^2}$ Horizontal traces are hyperbolas. Vertical traces are parabolas. The case where $c < 0$ is illustrated.	Hyperboloid of Two Sheets 	$-\frac{x^2}{a^2} - \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$ Horizontal traces in $z = k$ are ellipses if $k > c$ or $k < -c$. Vertical traces are hyperbolas. The two minus signs indicate two sheets.

- Cylindrical Coordinate system: Points are represented by (r, θ, z) . r, θ are polar coordinates of P onto the xy -plane and z is the usual z -coordinate in the Cartesian coordinate system.



To convert from cylindrical to rectangular:

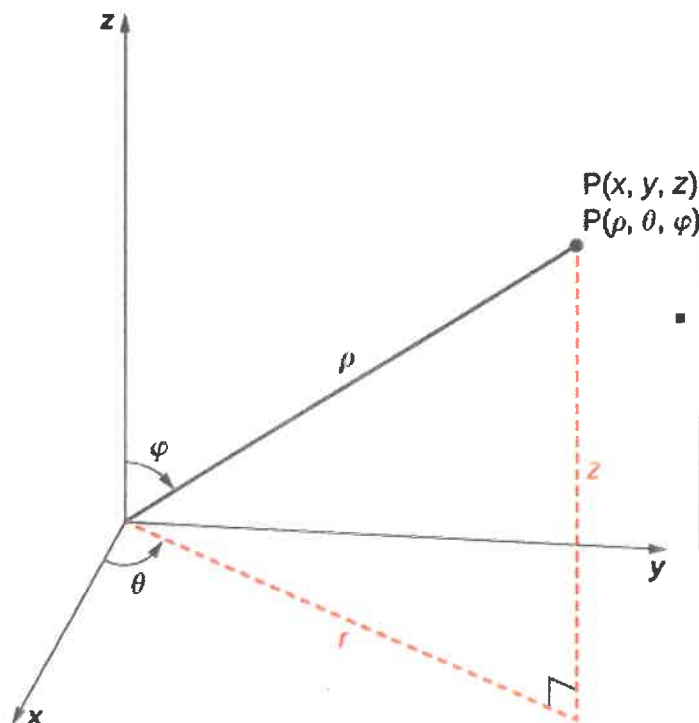
$$x = r \cos \theta \quad y = r \sin \theta \quad z = z$$

To convert from rectangular to cylindrical:

$$r^2 = x^2 + y^2 \quad \tan \theta = \frac{y}{x} \quad z = z$$

- NOTE: r can be negative.

- Spherical Coordinate system: Points are represented by (ρ, θ, ϕ) .
 ρ (Greek letter rho) = $|OP|$, the distance from the origin to P . $\rho \geq 0$
 θ is the same as in cylindrical.
 ϕ (Greek letter phi) is the angle between the positive z -axis and line segment $\overline{OP}$. $0 \leq \phi \leq \pi$



To convert from spherical to rectangular coordinates:

$$x = \rho \sin \phi \cos \theta \quad y = \rho \sin \phi \sin \theta \quad z = \rho \cos \phi$$

- Recall: $x = r \cos \theta$ and $y = r \sin \theta$.

To convert from rectangular to spherical:

$$\rho^2 = x^2 + y^2 + z^2 \quad \tan \theta = \frac{y}{x} \quad \phi = \arccos \left(\frac{z}{\sqrt{x^2 + y^2 + z^2}} \right)$$

- Spherical coordinates are useful in problems involving symmetry about a point. Put the origin at this point.